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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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TECHNICAL PRESENTATION

From Caesar to Vigenère: Evolution of Substitution Ciphers

Cryptography and Network Security (CSA5112)

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What Are Substitution Ciphers?



Definition

A substitution cipher replaces each letter of plaintext with another letter, symbol, or number based on a fixed rule or key.

Key Characteristics

- One of the oldest encryption techniques
- Uses a key for both encryption and decryption
- Provides message confidentiality
- Foundation for modern cryptography





Caesar Cipher

Developed by Julius Caesar around **50 BC** for secure military communication. Each letter is shifted forward by a fixed number of positions in the alphabet.

How It Works

Shift = 3: A → D, B → E, C → F...

HELLO → KHOOR

Advantages

- Simple to understand and implement
- Fast encryption and decryption

Limitations

- Only 25 possible keys
- Easily broken by brute force





Monoalphabetic Cipher

Rather than a fixed shift, each letter maps to a unique sub from a scrambled alphabet — dramatically expanding the

Example Substitution

Plain: ABCDEFGHIJKLMNOPQRSTUVWXYZ

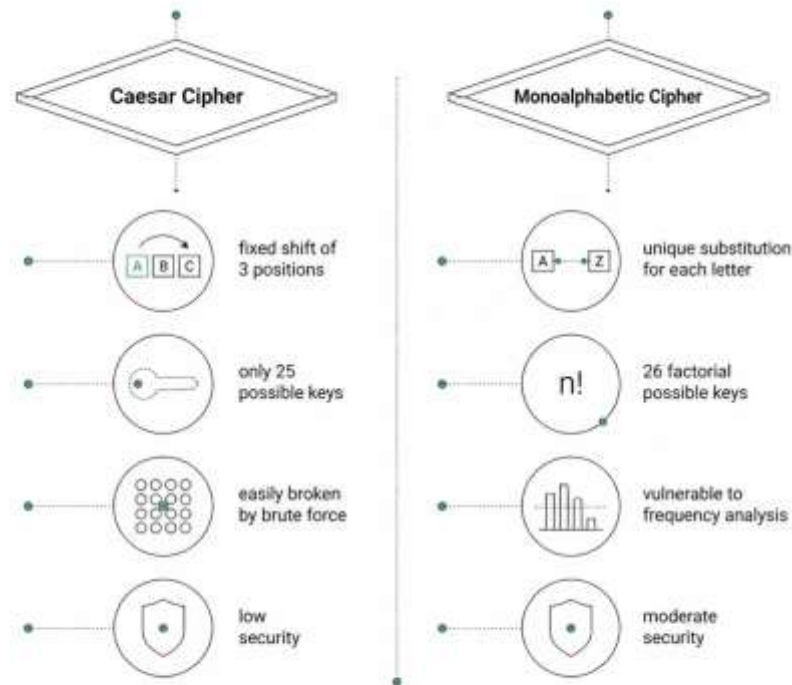
Cipher: QWERTYUIOPASDFGHJKLZXCVBNM

Advantages

- Vastly larger key space than Caesar
- Brute force attacks become impractical

Limitations

- Vulnerable to **frequency analysis**
- Letter patterns remain statistically visible





Vigenère Cipher

POLYALPHABETIC SUBSTITUTION

Invented by Blaise de Vigenère in the 16th century. A keyword determines which shift to apply at each position — multiple alphabets in one cipher.

Working Principle

Each letter of the keyword specifies a different Caesar shift. The keyword repeats across the message length.

Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E
Ciphertext	R	I	J	V	S

Why It's Stronger

Multiple Shift Patterns

The same plaintext letter encrypts differently depending on its position.

Resists Frequency Analysis

Letter frequencies are flattened across the ciphertext.

Very Large Key Space

Keyword length and content create enormous combinations.



Comparison of Ciphers

Each cipher represents a leap in cryptographic strength — more complex keys, larger key spaces, and better resistance to attack.

Feature	Caesar Cipher	Monoalphabetic	Vigenère Cipher
Key Type	Fixed Shift	Substitution Table	Keyword
Complexity	Low	Medium	High
Security Level	Low	Moderate	High
Frequency Analysis	Poor resistance	Poor resistance	Better resistance
Key Space	25 keys	$26! \approx 4 \times 10^{26}$	Very Large

Caesar Cipher

Monoalphabetic

Vigenère Cipher

Security improved at each stage by increasing key complexity and reducing statistical predictability.



Conclusion & Key Takeaways

01

Caesar Cipher

Introduced substitution encryption with a fixed shift — simple but historically significant.

02

Monoalphabetic Cipher

Expanded key complexity with unique letter mappings, though frequency analysis remained a threat.

03

Vigenère Cipher

Introduced polyalphabetic substitution via a keyword, resisting simple statistical attacks.

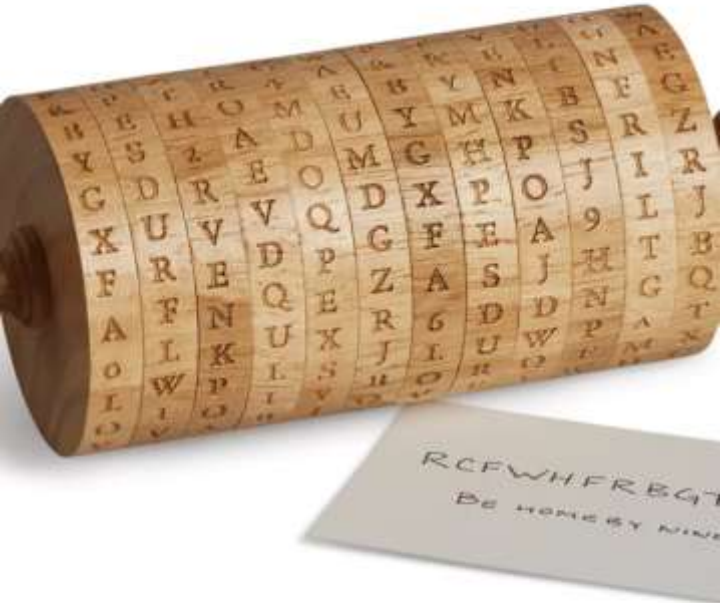
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Foundation for Modern Crypto

These ciphers established principles that underpin today's cryptographic systems.



Key Takeaway: As cryptographic techniques evolved, security improved by increasing key complexity and reducing predictability.



THANK YOU